Comparison of Semantic Organization in a Concept Feature Norm of 6–9-Year-Old Children with Thematic and Taxonomic Language Models

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Introduction

- **Concepts** are **mental** representations defined by observable and latent semantic features that enable categorization [1, 2].
- In adults, concepts are suggested to be organized **taxonomically** based on pre-existing categories for efficient retrieval (Figure 1.) [3].
- In contrast, in children **before the age of seven**, concepts are suggested to be organized primarily based on **thematic** relations defining situational co-occurrence (Figure 2.) [4, 5].
- Eliciting verbal descriptions of concepts from participants can be quantified into **semantic feature norms** to model inter-concept relations [6, 7].
  - Our **goal** is to **create a concept feature norm** based on Hungarian children producing concept definitions.

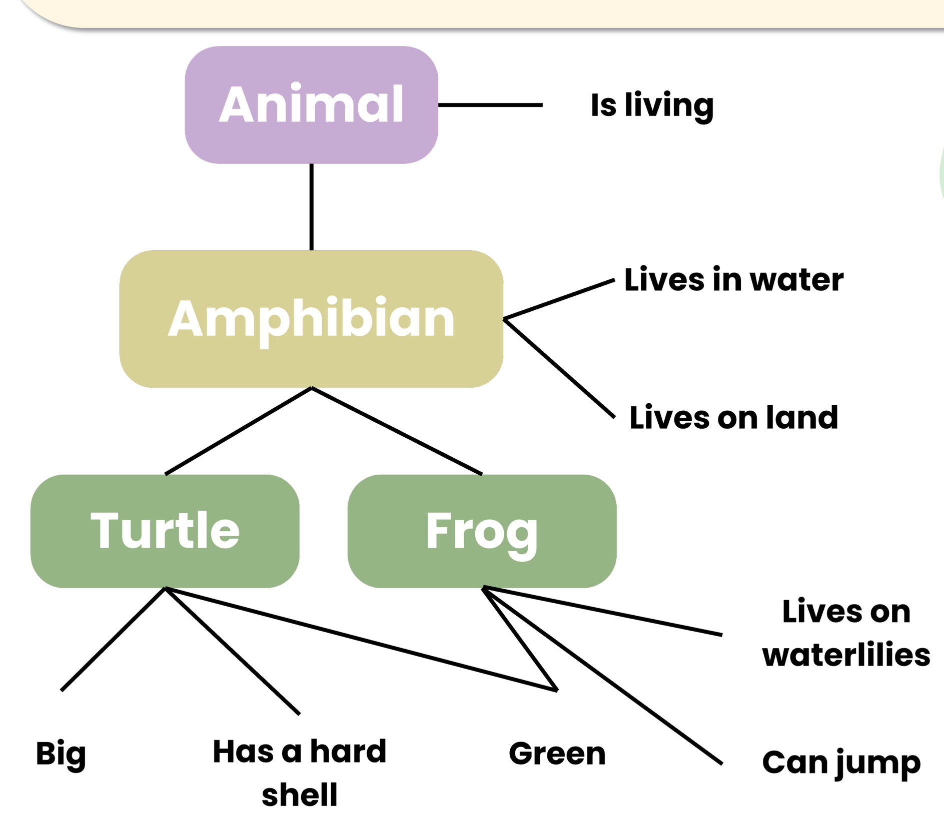

Figure 1. Taxonomic organization of two random concepts and their semantic features in our data. General attributes shared across subordinate concepts are stored at the superordinate level (e.g., "amphibian – lives in water"), while unique features are represented at the subordinate level (e.g., "turtle – has a hard shell").

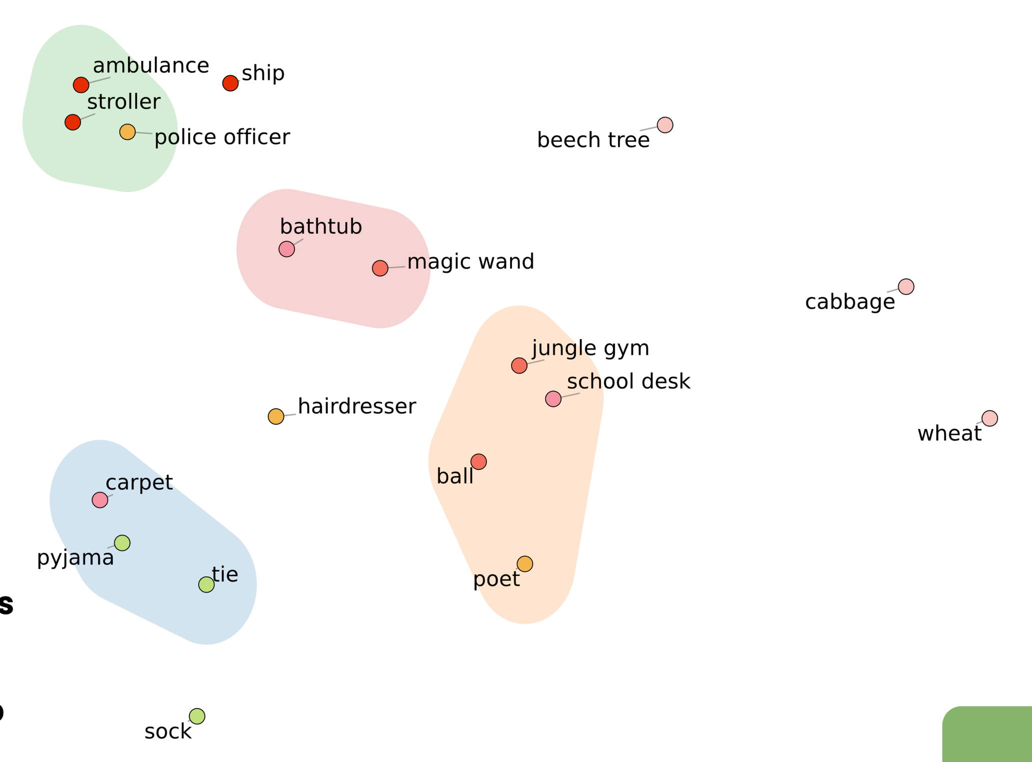

Figure 2. Thematic organization of 30 random concepts from our data, derived by aggregating shared semantic features via tf-IDF weighting, with dimensionality reduction applied using UMAP and cluster identification via DBSCAN. Colored dots represent a priori taxonomic categories. Clusters reflect contextual similarity rather than taxonomic membership (e.g., "ambulance" and "police officer" are part of different taxonomic orders, yet they might appear in similar contexts).

Questions

Does comparing verbal definitions of concepts by 6-9-year-olds to large language corpora reveal an underlying thematic representational structure?

H1: When modelling concept representations based on co-occurring verbal feature descriptions by 6-9-year-olds, we expect that the pairwise similarities will correlate positively with pairwise similarities based on a word-embedding derived from word-to-word co-occurrences, and not with pairwise similarities based on a priori encyclopaedic categories derived from WordNet.

Furthermore, we aim to develop a **methodological framework** for extracting semantic features and mapping children's conceptual representations from verbal descriptions.

Methods

- We selected **300 Hungarian concepts** denoting common, familiar concepts from 10 a priori **taxonomic** categories (Figure 5.) based on the *WordNet* lexical database [8].
  - Subsequently, we used a Hungarian *word2vec* word-embedding [9] to extract 200-dimensional vectors, with vector similarities representing **thematic** relations.
- We recruited 245 Hungarian children (130 females, M_{age} = 7.76, SD_{age} = 1.46) from fifteen elementary schools in Budapest and its agglomeration (Figure 4.), asking them to teach concept definitions to ***'Mimó'***, an imaginary character, via a **web-based application** (Figures 3-4).
  - We collected an average of 7.4 responses per concept (SD = 2.4, range = 1-25), with participants generating 9.54 features per concept on average (SD = 2.58, range = 4.5-17.0).
- Voice recordings were **manually transcribed** via a reliable protocol by 15 transcribers (mean pairwise Cohen's kappa = .94, SD = .02, all above .88).
  - Inter-rater reliability was assessed at the word level via a modified **Word Error Rate** approach [10], measuring pairwise comparisons between transcribers.

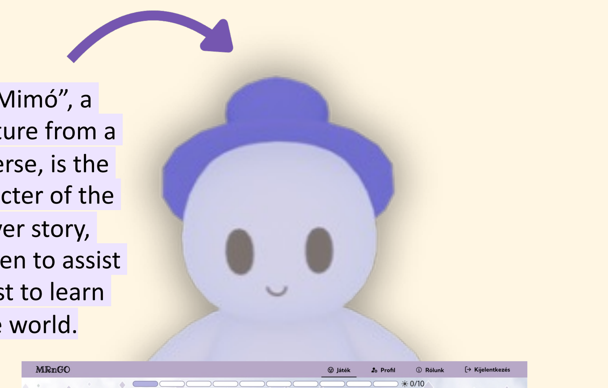

Figure 3. "Mimó", a magical creature from a distant universe, is the central character of the study's cover story, inviting children to assist in their quest to learn about the world.

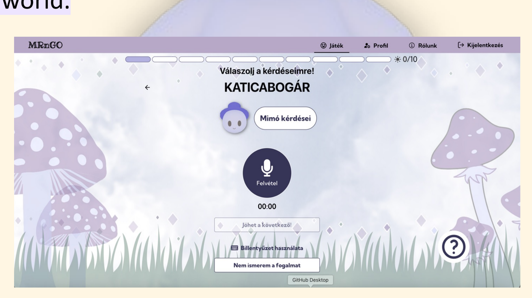

Figure 4. Main screen of the web application displaying the practice concept "ladybug", through which children are familiarized with the task and guided towards producing direct semantic features rather than indirect associative content.

Preprocessing

- We first **extracted tokens** from utterances via the Hungarian model of ***HuSpaCy*** [11] and retained tokens with **content words** (verbs, nouns, adverbs, adjectives, numerals, pronouns, adpositions, and postpositions) **filtering minimal responses** (e.g., 'yes' or 'no'), **comments** (e.g., 'I know'), and naming the concepts themselves.
- Tokens were **lemmatized** and compiled into **feature segments** (e.g., "have_four_leg", "live_water"), **retaining** specific **prefixes** (e.g., negation as 'NEG', 'it has' as 'HAS').
- Punctuations, conjunctions, stopwords and low-information tokens (e.g., "the") were either used as segment boundaries or discarded (Figure 6.).
- **FastText** vectors [12] were averaged across lemmas within each segment and submitted to **k-means clustering**, producing **300-dimensional clusters** (Figure 12) representing the proportional feature loading of each concept.

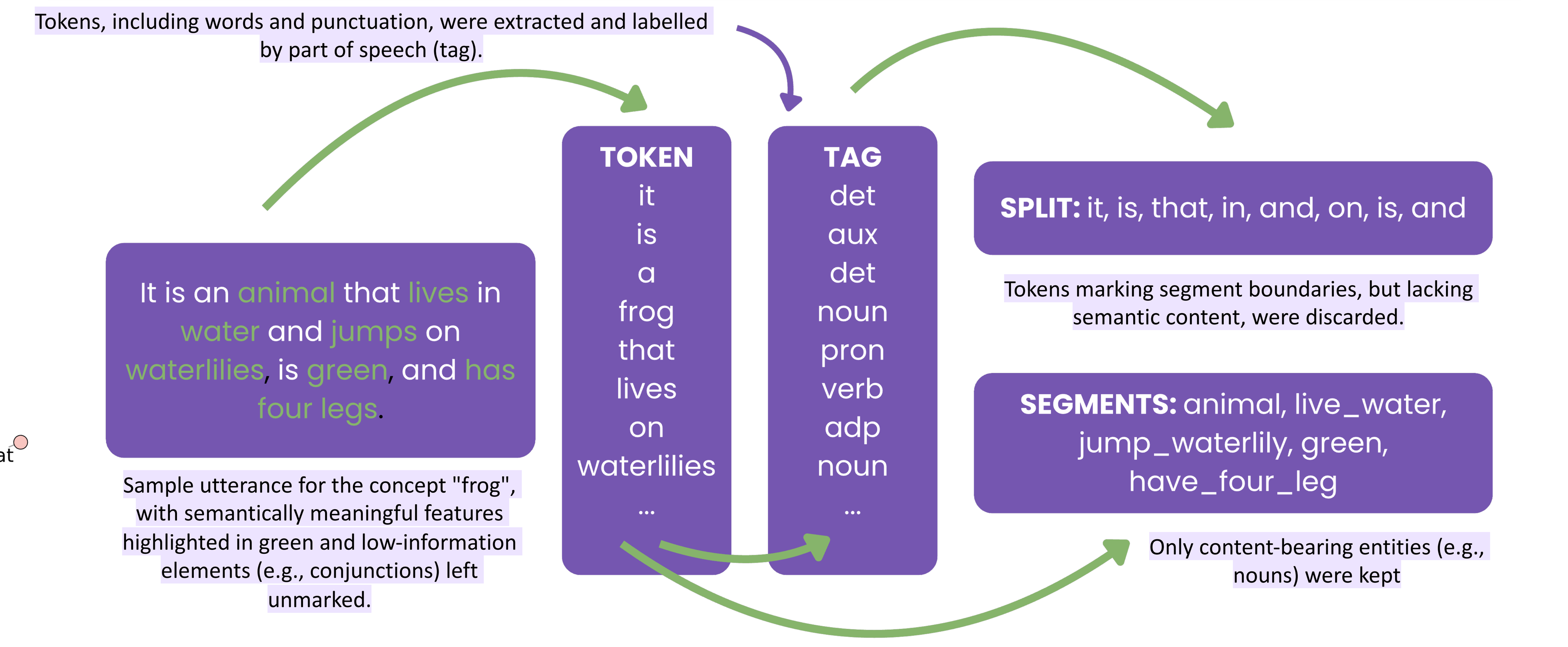

Figure 6. Feature extraction procedure through an example in our data.

Results

- We constructed **representational dissimilarity matrices** (RDM) via pairwise Pearson correlations between concept feature vectors, pairwise category agreements in a priori **taxonomic** model, and pairwise similarities between word2vec vectors reflecting a **thematic** model (Figure 7-9.). Then, we used **Spearman correlation** to compare similarities between the model RDMs and our data RDM.
- **Comparing** the **RDM** of **our data** with the **taxonomic RDM** yielded a **significant positive** result **(r = 0.18, p < .001)** (Figure 10.)
- **Comparing the RDM** of **out data** with the **thematic RDM** yielded **a significant positive result** as well **(r = 0.23, p < .001)** (Figure 11.)
- While **both** thematic and taxonomic structures **positively correlated** with our dataset, **thematic structure** showed **stronger alignment. Weak correlations** with both RDMs, however, indicate limited feature overlap across conceptual categories.

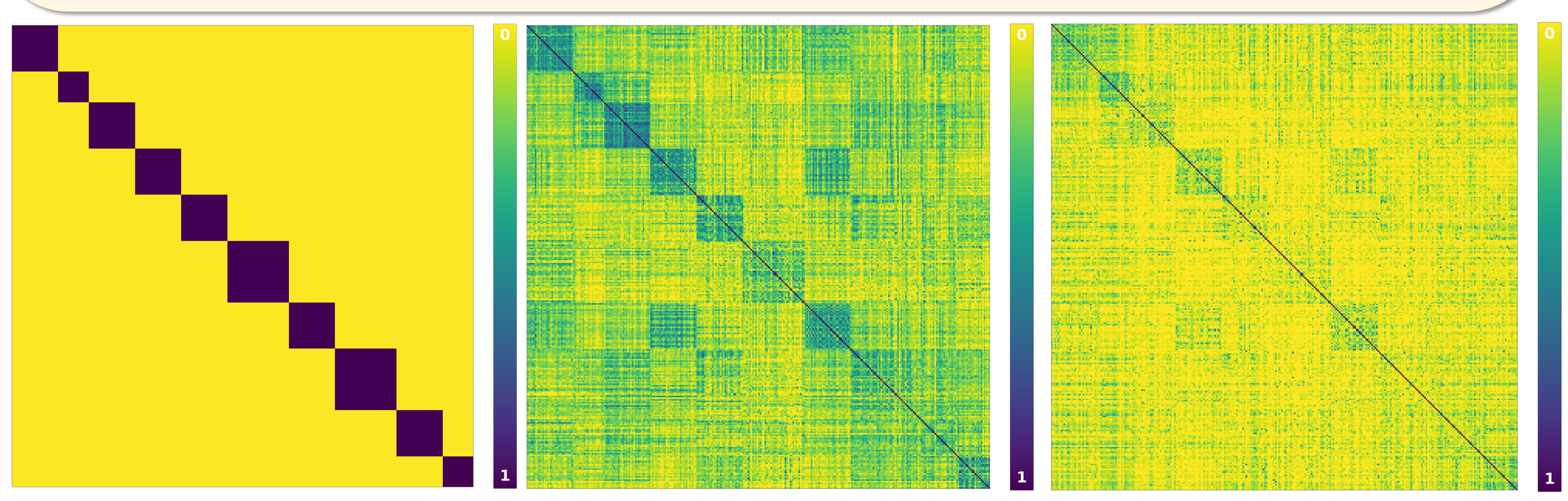

Figure 7-9. Representational Dissimilarity Matrices (RDMs) comparing pairwise similarities between: (1) the a priori taxonomic structure of 300 concepts derived from WordNet (2) a word2vec-derived thematic structure representing distance in vector-space of the 300 concepts (3) semantic similarity between concept vectors based on features extracted from children's verbal responses.

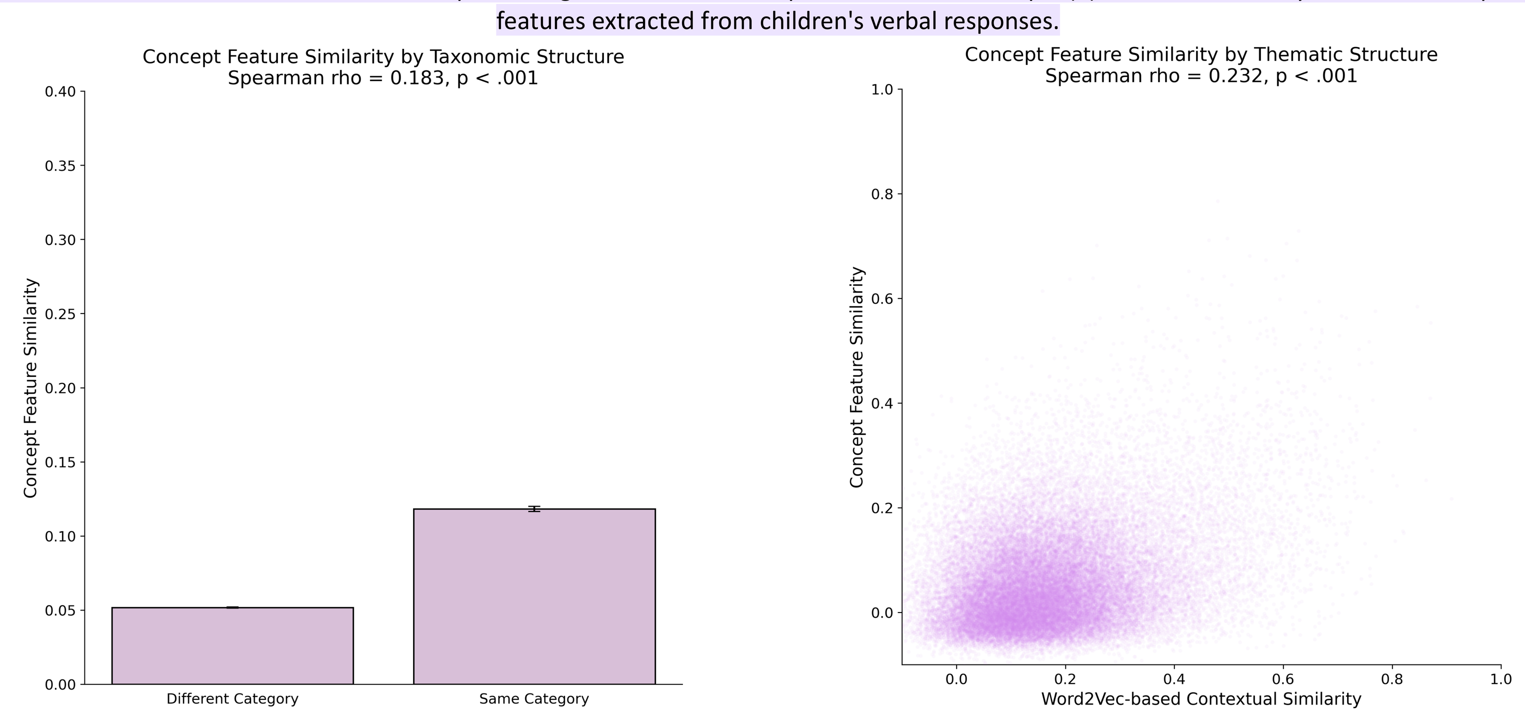

Figure 10-11. Pairwise correlation plots contrasting the dataset against a priori taxonomic and thematic structures. (1) Bar plot: within – (e.g., dog-duck) versus cross-category (e.g., dog-apple) similarity; (2) Scatter plot: contextual co-occurrence (x-axis) against shared semantic features per concept pair (y-axis).

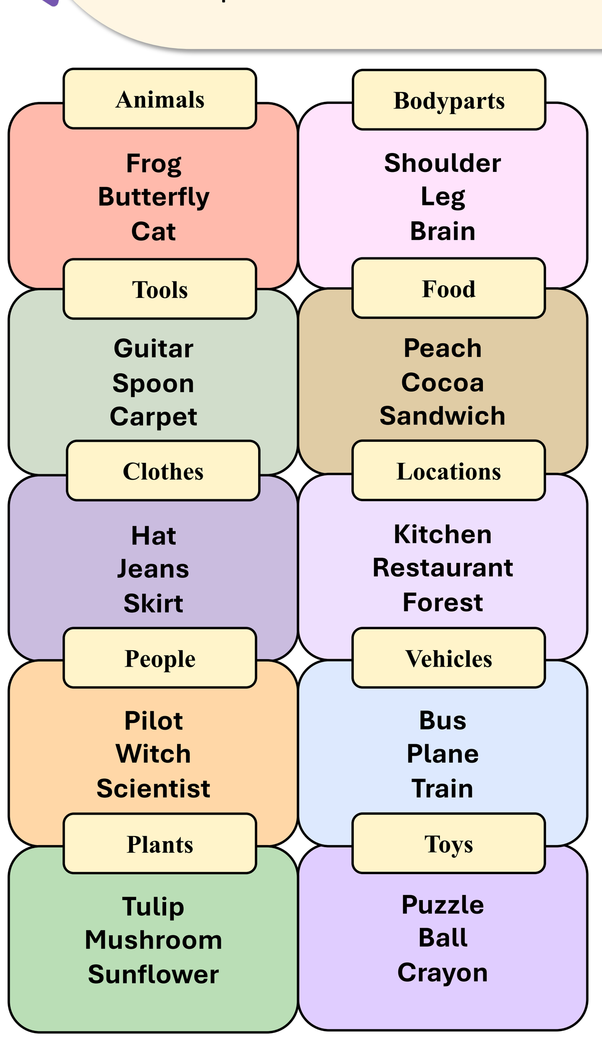

Figure 5. Sample concepts across the 10 taxonomic categories.

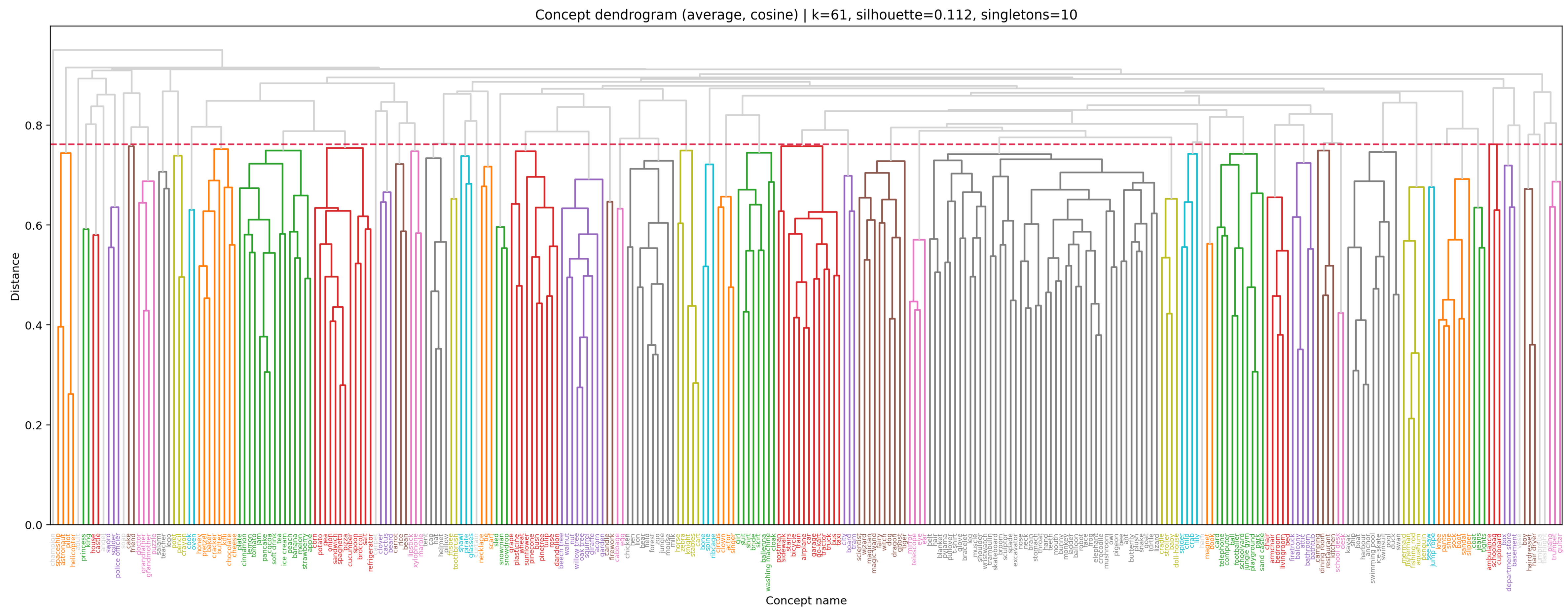

Figure 12. Hierarchical agglomerative clustering of concepts based on shared feature segments, with lower values indicating greater similarity. Colors represent 61 independent clusters (below red line). A silhouette score of 0.112 indicates fuzzy cluster boundaries, with singletons reflecting single-concept clusters.

Outstanding Questions

- ***How to optimize extraction of semantic features from free verbal productions?***
- ***What are the inferential limits of children's verbal descriptions of concepts?***
  - E.g., do negations signal the absence of a feature, or can they be interpreted as implying a feature of opposite meaning?
- ***How to optimize clustering via quantified feature relations?***
  - E.g., k-means, DBSCAN, agglomerative clustering etc.

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